Decoupling Language-Model Judgement from Constrained Portfolio Optimisation: A Measurement Protocol and Three Null Results

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Abstract—Language models are increasingly proposed as portfolio allocators. We argue that the interesting question is not whether such a model can produce weights, but whether it should, and we give an architecture and a measurement protocol that make the question answerable. The model's only output is a bounded confidence score that tilts expected returns inside a convex program; the solver, which alone sees the covariance structure, the caps and the turnover penalty, produces every weight. To test what that separation is worth, we run four arms over thirteen years on the same eligible universe: the model with company names, the model with names removed, a deterministic factor control with no model, and a monolithic arm in which the model returns weights directly. All three effects are indistinguishable from zero: temporal contamination +0.38 pp/yr ($p = 0.91$), value added over the deterministic control −0.05 pp/yr ($p = 0.99$), and the value of decoupling +0.81 pp/yr ($p = 0.78$). The monolithic arm breached no ceiling, but returned weights that failed to sum to one in five of thirteen years and silently omitted eligible names in two. Because null results are only informative when the measuring instrument is sound, we describe the evaluation infrastructure in equal detail: a survivorship-free panel rebuilt from the exchange's own quotation archive (514 issuers, 166 of which stop trading), receipt-date gating of regulatory filings, a nested selection protocol whose hindsight premium is measured at 0.65 pp/yr, and a deflated-Sharpe probability of 0.986 over 36 trials. We report seven defects found by auditing our own backtest, all of which had inflated the result.

Index Terms—portfolio optimisation, large language models, backtest overfitting, deflated Sharpe ratio, survivorship bias, temporally available data, emerging markets

I. Introduction

A language model asked for portfolio weights will produce them. Whether it should be asked is a separate question, and the literature has largely skipped it: papers that report a model beating a benchmark rarely include the control that would distinguish the model's judgement from the ranking signal it was given, and rarely test whether the model profited from recognising companies whose subsequent fate is in its training corpus.

This paper takes the opposite route. We fix an architecture in which the model cannot set a weight, then measure what that restriction costs. The measurement requires three comparisons that are usually conflated:

1) Temporal contamination: does the model do better when it can see company names than when it sees only anonymised financial facts? A positive, significant gap means it is remembering, not reasoning.
2) Added value: does the model beat a deterministic control that ranks the same universe by the same pre-declared factor? Without this arm, any result merely restates the factor.
3) Value of decoupling: does routing a bounded score through a convex solver beat letting the model emit weights directly? This is what the architecture is for.

Our contribution is threefold. First, the decoupled architecture and the protocol that makes these three questions separately answerable. Second, three null results on Brazilian equities over thirteen years, reported with the power caveats they deserve. Third, and inseparably, the evaluation infrastructure: a null result is only informative if the instrument would have detected an effect, so we document the panel construction, availability gating, the multiple-testing correction and the defects we found in our own work. Upon acceptance, we will deposit the evaluation artefacts and a versioned manifest in a permanent repository; during double-anonymous review, the same materials are available in an anonymised supplementary archive.

II. Related work

Markowitz's mean–variance formulation [1] remains the allocation substrate, with estimation error, concentration and turnover treated as first-class constraints rather than afterthoughts. Black and Litterman [2] established the pattern we follow structurally: qualitative views are combined with a mechanical allocator through a defined interface rather than replacing it. DeMiguel et al. [3] show that optimised portfolios frequently fail to dominate naive diversification out of sample, which motivates explicit caps and a frozen comparator. Novy-Marx and Velikov [4] show that frictions can erase anomaly profitability, which is why cost and tax are inside the evaluated protocol here rather than appended to it.

On inference, Bailey and López de Prado [5] give the deflated Sharpe ratio, which corrects the observed statistic for the number of trials and the higher moments of the return distribution; Bailey et al. [6] give the probability of backtest overfitting. Harvey et al. [7] document how weak the standard significance bar is once the size of the search is acknowledged. These are not optional refinements for a paper that evaluates 36 configurations.

Financial language models have been studied as domain models, research assistants and direct stock selectors.

BloombergGPT demonstrates the value of finance-specific corpora for general financial tasks [8], while FinGPT argues for an open, data-centric pipeline [9]. Direct market applications report that news-conditioned language-model scores can predict returns [10] and that live ChatGPT ratings co-move with subsequent earnings and returns [11]. Those results motivate our experiment but do not answer its architectural question. A model may appear predictive because it was handed an effective structured signal, because names reveal outcomes embedded in its training data, or because invalid allocations are repaired off-record. Named, anonymised, deterministic and monolithic arms are required to distinguish these mechanisms.

III. Architecture

A. The interface

Let $\mu \in \mathbb{R}^n$ be the expected-return vector estimated from information available at the decision date, Σ the covariance estimate, and w_{prev} the incumbent book. The model returns a confidence vector $c \in [-1, +1]^n$, and the only channel through which it reaches the portfolio is

$$\mu_i \leftarrow \mu_i \cdot (1 + \lambda c_i), \qquad c_i \in [-1, +1]. \tag{1}$$

The allocation is then the solution of

$$\max_{w} \; \mu^\top w - \frac{\gamma}{2} w^\top \Sigma w - \kappa \|w - w_{\text{prev}}\|_1 \tag{2}$$

subject to $\sum_i w_i = 1$, $w \geq 0$, an asset-class ceiling, and a per-issuer ceiling. The constants λ, γ and κ are fixed before the run and are not tuned per period.

One clarification matters for reproducibility, because the two are easy to conflate. The solver above is the allocator for every arm in the language-model experiment. The eleven-year published strategy uses a simpler allocator on the same screen: a hard top-N cut on the factor score, one share class per issuer, with weights assigned in proportion to score inside the same ceilings. We report both because the paper's claims about the model concern the first and the claims about realised returns concern the second, and a reader entitled to check either should not have to guess which is which.

Two properties follow, and they are properties of the type system rather than of the prompt. The model's output is a scalar in a closed interval, so there is no syntactic route by which it can express a weight, relax a ceiling, or admit an ineligible asset. And the solver is the only component that sees Σ, the caps and the turnover penalty, so the trade-off between expected return and risk is made where those quantities exist.

B. Pipeline

Each January the system executes five stages in a fixed order. Stages one to three are deterministic, the model enters only at stage four, and stage five is human.

(i) Dated universe. The universe of record is the exchange's own historical quotation archive [12], not a vendor feed. (ii) Available-at-decision fundamentals. Regulatory filings [15], [16] carry both a reference date and a receipt date; only filings received on or before the decision date may enter. (iii) Screen. Assets must clear liquidity, profitability, return on capital and solvency, with operating companies and financial institutions screened on different quantities because their standardised statements are not comparable. (iv) Ranking and allocation. Survivors are ranked and the ranking becomes the confidence vector; allocation is by the solver in the experimental arms and by the score-proportional top-N rule in the published strategy, in both cases within the ceilings and with the residual held in the domestic overnight rate [17]. (v) Human review. The output is a proposal carrying the applied policy, the screen with its rejection reasons, the weights, order sizes with lot rounding and participation in average traded volume, and the estimated cost. No order is transmitted.

The economic hypothesis is fundamental-led rather than language-model-led. Quality and value are the primary accounting dimensions, twelve-month momentum is a market confirmation signal, and liquidity is an execution gate. Banks and operating companies use different accounting quantities. The model does not create this signal: in production architecture it converts approved facts into thesis, risk and review questions. MVO is an independent benchmark in the published record and an allocator only inside the four-arm experiment.

IV. Evaluation infrastructure

A null result is a claim about an instrument as much as about a treatment. This section describes the instrument.

A. Survivorship

A public adjusted-price provider serves the companies that still exist, so a universe built from one silently deletes every company that was delisted, acquired or wound up. Rebuilding from the exchange archive over 2010–2025 yields 4,066 sessions and 514 issuers, of which 166 stop trading before December 2025; 77 of those disappear before 2020, and 58 had already fallen more than 60% from their own peak when they stopped trading. These are precisely the worst returns in the sample.

Delisted tickers are not served by any provider, so corporate actions must be recovered from round, persistent price ratios in the archive itself. We validate the detector against an archived provider event feed on the subset where one exists and publish both sides: precision 88.1% (340 of 386 applied adjustments coincide with a confirmed event) and recall 23.3% (103 of 443 provider events detected). The low recall is a real limitation and is stated as one; the detector is deliberately conservative and prefers not to adjust over adjusting wrongly.

B. Availability gating

Trailing-twelve-month quantities are bridged as annual plus current year-to-date minus comparative year-to-date, using the year-to-date column only. The interim filing also publishes the isolated quarter, and mixing the two silently corrupts every ratio from the second quarter onward. This is the kind of defect that produces a better backtest and no error message.

C. Holding, cost and tax

The book is bought in January and held. This is worth stating because the natural vectorised expression of a backtest, compounding the inner product of returns and weights, silently rebalances to fixed weights every session at zero cost, which is neither the protocol under evaluation nor anything an investor can execute. Costs are the published exchange fee plus a slippage term growing with the position's participation in average traded volume. Taxation follows the domestic regime, with gains taxed when a review actually realises them and the final year charged as a full liquidation.

D. Choosing the rule without reading the answer

Our first protocol failed this test, and the failure is instructive. A grid of candidate rules was evaluated, the best was published, and its rank moved from 57th on the declared training criterion to 1st on the holdout, which is only possible if the holdout was read.

The current protocol is nested. For decision year t, the 36 configurations are ranked using only years that closed before t, by Sharpe of the excess return over cash; year t is then evaluated once. Changing configuration is charged a full-turnover rebalance. Ranking requires three closed years, so 2012–2014 is the selection window and 2015–2025 is the evaluation window, giving eleven decisions.

Three quantities are published with every result: the nested outcome, the full-sample winner, and the difference between them. That difference, the hindsight premium, is what an author would gain by publishing the search winner as though it were attainable. On this panel it is 0.65 pp/yr. On a panel with one year less of training it was 4.98 pp/yr, which is a useful calibration of how much the measure moves with sample length.

With 36 trials, the winner's Sharpe is biased upward by the search itself. The observed excess-return Sharpe is 0.933 against an expected maximum under the null of 0.355, giving a deflated-Sharpe probability of 0.986, above the 95% threshold.

V. Experiment

A. Design

Four arms over thirteen years, same eligible universe, same solver, same caps:

- Named: the model sees company names and returns a bounded score.
- Anonymised: identical, but issuers are identified only by integers.
- Deterministic: control with no model, ranking by the pre-declared factor score.
- Monolithic: the counterfactual in which the model returns weights directly, with no solver and no constraints.

Comparisons are paired by year and tested with a paired t-test.

TABLE I
Four arms, thirteen years, same eligible universe.

Arm	CAGR	Win yrs	Eq. wt.
Named	13.00%	7 of 13	43.8%
Anonymised	12.62%	10 of 13	44.7%
Deterministic	12.67%	8 of 13	52.2%
Monolithic	11.81%	7 of 13	55.0%
Cash (CDI)	9.59%	—	—

TABLE II
The three effects. None survives testing.

Effect	Annualised	p
Contamination (named − anonymised)	+0.38 pp	0.91
Added value (anonymised − deterministic)	−0.05 pp	0.99
Decoupling (anonymised − monolithic)	+0.81 pp	0.78

B. Results

Table II reports three nulls, and each means something different.

No detectable contamination. The named arm did not profit from recognising companies whose fate the model could plausibly know from training. This was the most serious objection to any use of a language model on historical data, and it did not materialise in this configuration. We do not claim contamination is absent in general; we claim it was not detected here, at this model's training cutoff.

No detectable added value. The model reproduces the deterministic ranking rather than adding judgement. The gap is five hundredths of a percentage point, with a negative sign. This is an uncomfortable finding for the prevailing narrative about language models in investment management, which is a reason to publish it rather than a reason not to.

Keeping the model away from the weights did not cost performance. The monolithic arm breached no ceiling in thirteen years, a point in its favour that we report because the opposite would have been the convenient finding. What it did produce was arithmetically invalid: weights failing to sum to one in five of thirteen years, ranging from 0.92 to 1.017, and 65 and 83 eligible names silently omitted in 2021 and 2022. The argument for the solver is therefore not that it prevents a limit breach. It is that it returns an allocation requiring no repair, and that repair on a malformed weight vector is itself an unlogged decision.

Thirteen annual observations make these underpowered tests. The honest statement is that no effect was detected, not that none exists.

TABLE III
Defects, all of which had inflated the result.

Defect	Correction
MVO comparator identical to the strategy	Independent implementation
Vendor panel dropped delisted issuers	Universe rebuilt from the exchange archive
Delisting year discarded entirely	Position liquidated into cash
No tax modelled	Domestic tax regime
Flat cost ignoring liquidity	Slippage by participation in volume
Implicit daily rebalancing	Exact buy-and-hold path
Session filter using a global peak	Local rolling peak

C. Robustness and the separate product record

Holding the rule and panel fixed, annual reselection compounded at 18.58% before costs, versus 16.37% quarterly and 13.92% monthly. After tax, the paired gaps were 1.55 pp/yr ($p = 0.38$) and 3.27 pp/yr ($p = 0.31$); no faster cadence improved the selection. A retention buffer also raised carryover from 36% to 42% and reduced one-year holdings from 21 to 16, but its 0.33 pp/yr return difference was not significant ($p = 0.14$). These are stability checks, not additional return claims.

The supplementary product record distinguishes the Ibovespa total-return index, which reinvests distributions in a theoretical portfolio [13], from BOVA11, an investable ETF subject to fees, trading costs and tracking difference [14]. A retrospective event-risk extension reduced maximum drawdown from 47.8% to 28.7%, but added no detectable return in 2019–2025 ($p = 0.964$). It was designed after Covid-19 and omits intrayear tax, so it is not prospective validation and does not enter the language-model tests.

VI. Defects found by auditing our own backtest

Seven defects were found and corrected. Every one inflated the result, which is what one should expect: an error that hurts your own numbers is noticed immediately and an error that helps them is not.

After the final correction the published CAGR fell and the daily drawdown rose from 30.4% to 47.8%. The figures were published as they came out.

VII. Limitations

The evaluation window is also the development window; the 0.65 pp hindsight premium bounds that problem without making the record prospective. The corporate-action detector's 23.3% recall still requires primary-record reconciliation, while issuers without provider coverage receive an explicitly listed cross-sectional median distribution yield. Eleven annual observations remain few: the deflated-Sharpe probability corrects search bias, not sample size. The model experiment covers one model and one training cutoff. The annual cadence beat quarterly and monthly reselection here, but is not universally optimal. Benevente 2 was designed after Covid-19 and omits intrayear tax, so it requires pre-registration and prospective evaluation. Timestamped news and event-driven decisions belong in a separate protocol. Factor transformations, thresholds, the configuration grid and annual choices are disclosed in the supplementary artefacts.

VIII. Conclusion

We measured what a language model contributes when it cannot set portfolio weights. Over thirteen years of Brazilian equities, nothing was detectable: names did not help, the model did not beat its deterministic factor, and keeping it away from the weights cost nothing. The dated screen generated the return. A bounded, typed interface therefore gives up nothing measurable while avoiding the malformed weights returned by the monolithic arm. The event-risk extension reaches the same architectural conclusion: a lagged deterministic control can reduce tail risk without asking a model to infer a crisis from prose. Claims for model discretion should be tested against both deterministic and anonymised controls.

Statement on generative AI

A generative-AI assistant was used for language editing and code review. The authors independently verified every calculation, citation and claim and accept full responsibility for the manuscript. The empirical results were generated by deterministic project code, not by the assistant [18].

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